SULIT

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA
MID-TERM EXAMINATION SEMESTER I 2020/2021

SUBJECT CODE	**: SCSR2043/SECR2043/SCSS4213**
SUBJECT NAME	**: OPERATING SYSTEM (PART 2)**
SECTION	**: 01/02/03**
TIME	**: 90 MUNITES**
DATE/DAY	**: 21 DECEMBER 2020**
VENUES	**: 8.50 PM – 10.20 PM**

INSTRUCTIONS:

There are TWO parts. Part 1 consists of 30 multiple choice questions and Part 2 consists of 21 multiple choice questions related to computation and critical thinking. This test will contribute 25% towards the total marks of 100%.

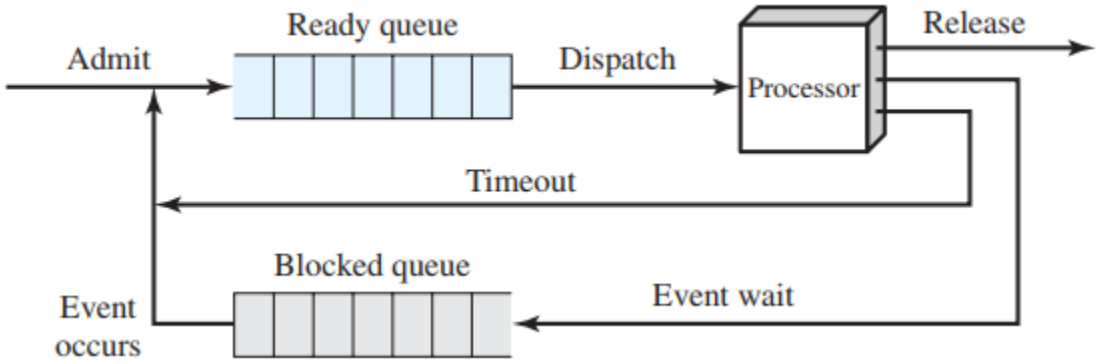
(Please Write Your Lecturer Name and Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section (Circle)	
Lecturer Name (Circle)	

This question paper consists of 16 printed pages excluding this page.

PART 2

1. Which one is **FALSE** about the figure below



I. As each process is admitted to the system, it is placed in the Ready queue

II. When a running process is removed from execution, it is either terminated or placed on the Ready queue or Blocked queue

III. When an event occurs, any process in the Blocked queue that has been waiting on that event only is moved to the Ready queue

IV. When it is time for the OS to choose another process to run, it selects one from the Ready queue

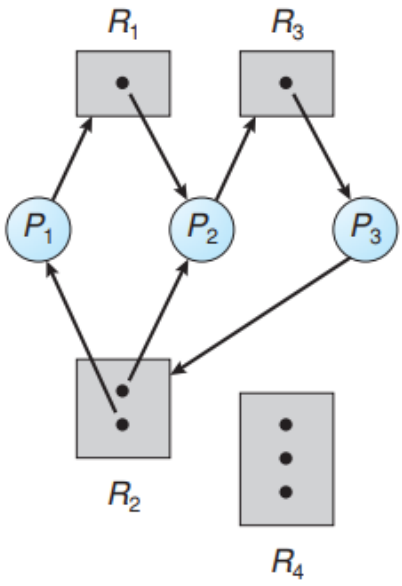
A. II, III and IV

B. I, II, III and IV

C. I, II, and III

D. I, III and IV

2. Which one is **TRUE** about the figure below



I. Process *P*3 is waiting for either process *P*1 or process *P*2 to release resource *R*2

II. Process *P*1 is waiting for process *P*2 to release resource *R*1

III. Processes *P*1, *P*2, and *P*3 are deadlocked

IV. There are two cycles in the resource-allocation graph

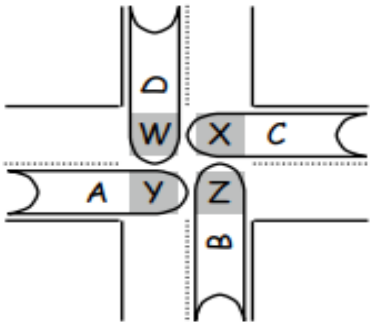
A. II, III and IV

B. I, II and III

C. I, III and IV

D. I, II, III and IV

3. If we assume that processes = vehicles and resource = road, which one is **FALSE** about the figure below?



Cars deadlocked
in an intersection

I. No preemption happens because W, X, Y and Z can only hold one car at a time

II. The processes are A, B, C and D while resources are W, X, Y and Z

III. A waits for B's Z, B waits for C's X, C waits for D's W and D waits for A's Y is why hold and wait happened in this traffic deadlock

IV. Circular wait happens in this traffic deadlock because A holds Y but waits for Z

A. II and IV

B. I, III and IV

C. II and III

D. II, III and IV

4. The following table illustrates 5 processes that are being scheduled using a preemptive **Priority scheduling** algorithm. Each process is assigned a numerical priority, with the smaller number indicating a higher relative priority. Which timeline chart illustrate the execution of the processes?

Process	Arrival Time	CPU Cycle	Priority
P1	0	9	4
P2	3	4	2
P3	4	1	3
P4	6	4	2
P5	8	2	1

A.

P1	P2	P2	P4	P5	P4	P3	P1
0	3	7	8	10	13	14	20

B.

P1	P2	P2	P4	P5	P4	P3	P1
0	3	7	8	10	13	14	20

C.

P1	P2	P2	P3	P5	P4	P1	
0	3	4	6	8	10	14	20

D.

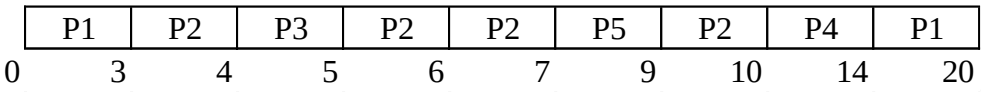
P1	P2	P2	P4	P5	P4	P3	P1
0	3	6	8	10	13	14	20

5. Referring to Question 4, what are the average waiting time (AWT) and average turnaround time (ATT)?

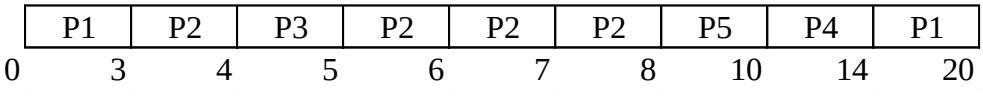
A. AWT = 4.6 and ATT = 8.6
B. AWT = 2.2 and ATT = 8
C. AWT = 3.8 and ATT = 7.8
D. AWT = 4.8 and ATT = 8.4

6. Referring to Question 4, what is the timeline chart if the execution of the processes is using SRT scheduling algorithms.

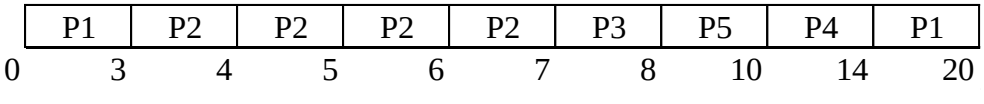
A.



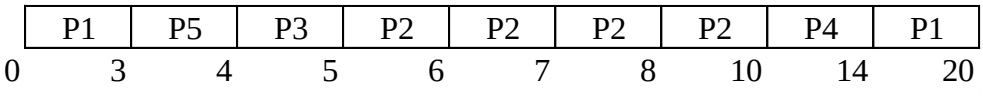
B.



C.



D.



7. Five processes P1, P2, P3, P4 and P5 arrived at time t_0 with different CPU bursts. Assume that the **Multilevel Feedback Queue (MFQ)** scheduling algorithm is applied with 4 queues (Q_i) where Q1, Q2 and Q3 using Round Robin with time quantum set to 6 CPU bursts, and the last Q level using FCFS, calculate the average turnaround time bad average waiting time if the burst time for each processes are P1 = 40, P2 = 5, P3 = 25, P4 = 20, and P5 = 6?

A. AWT = 39.3, ATT = 52.9

B. AWT = 40.6, ATT = 59.8

C. AWT = 36.1, ATT = 59.8

D. AWT = 40.6, ATT = 52.9

8. Five processes P1, P2, P3, P4 and P5 arrived at time t_0 with different CPU bursts. Assume that the **Multilevel Feedback Queue (MFQ)** scheduling algorithm is applied with 4 queues (Qi) where Q1, Q2 and Q3 using Round Robin with time quantum set to 6 CPU bursts, and the last Q4 is using FCFS, which process(es) will finish at level Q1 if the burst time for each processes are P1 = 40, P2 = 5, P3 = 25, P4 = 20, and P5 = 6?

A. No process

B. Process P1, P3 and P4

C. Process P2 and P3

D. Process P2 and P5

9. Five processes P1, P2, P3, P4 and P5 arrived at time t_0 with different CPU bursts. Assume that the **Multilevel Feedback Queue (MFQ)** scheduling algorithm is applied with 4 queues (Qi) where Q1, Q2 and Q3 using Round Robin with time quantum set to 6 CPU bursts, and the last Q4 is using FCFS, at what time the second process in Q1 enter to thee CPU if the burst time for each processes are P1 = 40, P2 = 5, P3 = 25, P4 = 20, and P5 = 6?

A. 29

B. 41

C. 35

D. 12

10. The figure below shows the implementation of Producer-Consumer using Semaphores. Supposed the initial values of the buffer size, **`N=10, mutex=1, empty=8`** and **`full=2`**. What are the final output if the following sequence of order is executed: consumer, consumer, producer?

Producer	**Consumer**

`do {`
`  produce an item`
`  …`
`  wait(empty);`
`  wait(mutex);`
`  …`
`  add item to buffer`
`  …`
`  signal(mutex);`
`  signal(full);`
`} while (TRUE);`
	`do {`
	`  …`
	`  wait(full);`
	`  wait(mutex);`
	`  …`
	`  remove an item from buffer`
	`  …`
	`  signal(mutex);`
	`  signal(empty);`
	
	`} while (TRUE);`

A.

Process	Wait(full)	Wait(mutex)	Signal (mutex)	Signal (empty)
Initially	2	1		8
Consumer	2	1	0	9
Consumer	0	0	1	10
Process	**Wait(empty)**	**Wait(mutex)**	**Signal (mutex)**	**Signal (full)**
Producer	10	0	0	3

B.

Process	Wait(full)	Wait(mutex)	Signal (mutex)	Signal (empty)
Initially	2	1		8
Consumer	1	0	1	9
Consumer	0	0	1	10
Process	**Wait(empty)**	**Wait(mutex)**	**Signal (mutex)**	**Signal (full)**
Producer	10	0	1	2

C.

Process	Wait(full)	Wait(mutex)	Signal (mutex)	Signal (empty)
Initially	2	1		8
Consumer	1	0	1	9
Consumer	0	0	1	10
Process	**Wait(empty)**	**Wait(mutex)**	**Signal (mutex)**	**Signal (full)**
Producer	9	0	1	1

D.

Process	Wait(full)	Wait(mutex)	Signal (mutex)	Signal (empty)
Initially	2	1		8
Consumer	2	0	2	9
Consumer	0	0	1	9
Process	**Wait(empty)**	**Wait(mutex)**	**Signal (mutex)**	**Signal (full)**
Producer	9	0	1	2

11. Given the following table, which timeline chart illustrate the execution of the processes using Round Robin (RR) scheduling algorithm with the time quantum set to 3.

Process	Arrival Time	CPU Cycle
P1	0	4
P2	1	6
P3	2	3
P4	3	5
P5	4	7

0 3 6 9 12 15 16 19 21 24 25

A.

	P1	P2	P3	P4	P5	P1	P3	P4	P5	P5
0	3	6	8	11	14	15	18	20	23	25

B.

	P1	P2	P3	P1	P4	P4	P5	P5	P2	P4
0	3	6	9	10	13	15	18	21	22	25

C.

	P1	P2	P3	P4	P1	P2	P4	P5	P5	P5
0	3	6	8	11	13	16	18	21	24	25

D.

	P1	P2	P3	P4	P5	P1	P2	P4	P5	P5
0	3	6	9	12	15	16	19	21	24	25

12. The following figure illustrates the Peterson's algorithm for a process. Consider two processes labelled as P0 and P1. If the `flag(0)= TRUE`, `flag(1)= TRUE`, and `turn = 0`, choose the **FALSE** statement(s) from the followings.

Process P0

```
While (TRUE)
{
   flag(0) = TRUE;
   turn = 1;
   while (flag(1) && turn == 1)
      ;  /* do nothing  */
   critical section
   flag(0) = FALSE;
   remainder section
}
```

I. *P1* can enter its critical section only when `flag(1) = FALSE`

II. *P0* is requesting to enter its critical section just after *P1*.

III. *P0* can proceed to enter its critical section.

IV. *P1* needs to wait to enter its critical section.

A. I, II and IV

B. I and II

C. I and III

D. All of the above

13. Peterson's Algorithm: Given a code for Process 0 and Process 1 in the figure below and initial status at time *t0* and event for *t1* and *t2* in the table below. What are the appropriate values for *flag* and *turn* at t_1 and t_2 after t_0, if Process 1 initiates the request to enter the critical section?

Process 0	Process 1
do { **flag[0]** = TRUE; **turn** = 1; while (**flag[1]** && **turn** == 1) ; /* do nothing */ /*critical section*/ **flag[0]** = FALSE; /*remainder section*/ } while (TRUE);	do { **flag[1]** = TRUE; **turn** = 0; while (**flag[0]** && **turn** == 0) ; /* do nothing */ /*critical section*/ **flag[1]** = FALSE; /*remainder section*/ } while (TRUE);

A.

time	flag[0]	flag[1]	Turn	Events
t0	FALSE	FALSE	1	-
t1	FALSE	TRUE	0	P1 request to enter CS1
t2	TRUE	FALSE	0	P0 requests to enter CS_0

B.

time	flag[0]	flag[1]	Turn	Events
t0	FALSE	FALSE	1	-
t1	FALSE	TRUE	0	P1 request to enter CS1
t2	TRUE	TRUE	1	P0 requests to enter CS0

C.

time	flag[0]	flag[1]	Turn	Events
t0	FALSE	FALSE	1	-
t1	FALSE	FALSE	0	P1 request to enter CS1
t2	TRUE	TRUE	1	P0 requests to enter CS0

D.

time	flag[0]	flag[1]	Turn	Events
t0	FALSE	FALSE	1	
t1	FALSE	TRUE	0	P1 request to enter CS1
t2	FALSE	TRUE	0	P0 requests to enter CS0

14. A system with 4 processes (P1, P2, P3, P4) are sharing 3 type of resources (A, B, C). The Banker's algorithm is used to deal with deadlocks state in the system. Gtven the table below compute the Need Matrix for each processes.

	Allocation			Maximum			Available		
	A	B	C	A	B	C	A	B	C
P1	1	2	0	5	5	3	2	4	3
P2	1	1	0	2	3	3			
P3	0	1	3	3	3	6			
P4	3	1	0	5	7	2			

A.

Need		
A	B	C
3	1	0
0	1	0
1	1	3
3	3	1

C.

Need		
A	B	C
1	2	3
1	3	3
2	3	0
1	3	3

B.

Need		
A	B	C
4	2	3
1	2	3
3	2	3
2	6	2

D.

Need		
A	B	C
6	7	3
3	4	3
3	4	9
8	8	2

15. Referring to Question 14, a system with 4 processes (P1, P2, P3, P4) are sharing 3 type of resources (A, B, C). The Banker's algorithm is used to deal with deadlocks state in the system. Given the table below, calculate the initial work for the Safety Algorithm.

A. (4, 2, 3)

B. (2, 4, 3)

C. (1, 2, 0)

D. (3, 5, 3)

16. Referring to Question 14, a system with 4 processes (P1, P2, P3, P4) are sharing 3 type of resources (A, B, C). The Banker's algorithm is used to deal with deadlocks state in the system. Given the table below, what are the new value of work for each processes after the execution?

A. P1= (1,5,3,2), P2= (3,14,11,7), P3= (2,14,11,7) and P4= (1,11,6,4)

B. P1= (0,0,0,0), P2= (1,7,5,5), P3= (1,0,0,3) and P4= (1,11,6,4)

C. P1= (0,0,0,0), P2= (3,14,11,7), P3= (1,3,5,3) and P4= (1,11,6,4)

D. P1= (1,5,3,2), P2= (1,7,5,5), P3= (1,3,5,3) and P4= (1,11,6,4)

17. Referring to Question 14, a system with 4 processes (P1, P2, P3, P4) are sharing 3 type of resources (A, B, C). The Banker's algorithm is used to deal with deadlocks state in the system. What are the safe state sequence after the execution?

A. P1 – P3 – P4 – P2

B. P2 – P3- P4- P1

C. P1 – P4 – P3 – P2

D. P1 – P2 – P4 – P3

18. Referring to Question 14, a system with 4 processes (P1, P2, P3, P4) are sharing 3 types of resources (A, B, C). The Banker's algorithm is used to deal with deadlocks state in the system. If the process P1 has been allocated with (1, 0, 0) of resources at first, and then requests more, 4 resources and 2 resources for type B and C, respectively. Choose the specific variable(s) of the Resource Request algorithm that must be updated for P1

I. Request

II. Need

III. Allocation

IV. Available

A. I, II, III and IV

B. I, II and III

C. II and III

D. I and IV

19. The following table shows three processes P1, P2 and P3 run concurrently in a Unix system. Assume that all the flags, A, B, and C are initially initialized to 0. What will the possible output be?

P0	P1	P2

`wait(B)`
`print("B ");`
`signal(C)`
	`signal(C)`
	`print("A ")`
	`signal(B)`
		`wait(C)`
		`print("C ")`

A.

Process	A	B	C	Output
Initial	0	0	0	-
P1	0	1	1	A
P0	0	0	2	B
P2	0	0	1	C

B.

Process	A	B	C	Output
Initial	0	0	0	-
P0	0	0	1	B
P2	0	1	2	A
P0	0	0	1	C

C.

Process	A	B	C	Output
Initial	0	0	0	-
P1	0	1	1	A
P2	0	1	0	C
P0	0	0	1	B

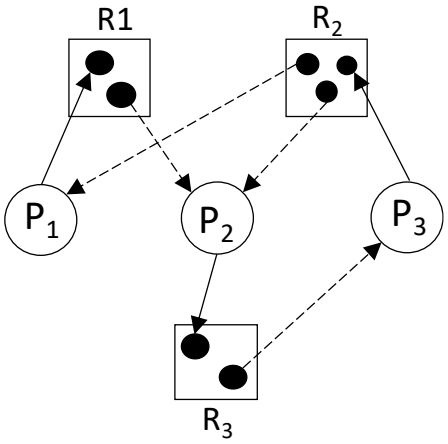
D.

Process	A	B	C	Output
Initial	0	0	0	-
P2	0	0	0	C
P1	0	1	1	A
P0	0	0	2	B

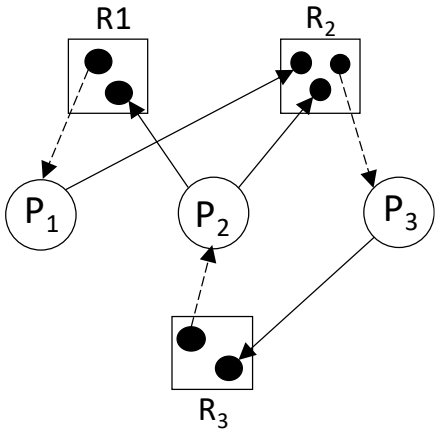
20. Which is the complete Resource-Allocation Graph (RAG) given the following set of edges that consists of request and assignment edges?

$$E = \{P_1 \rightarrow R_1, R_2 \rightarrow P_1, P_2 \rightarrow R_3, R_1 \rightarrow P_2, R_2 \rightarrow P_2, P_3 \rightarrow R_2, R_3 \rightarrow P_3\}$$

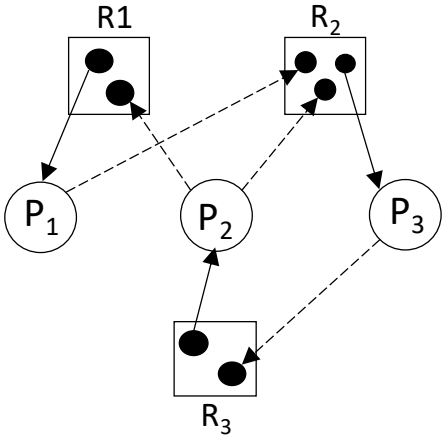
A.



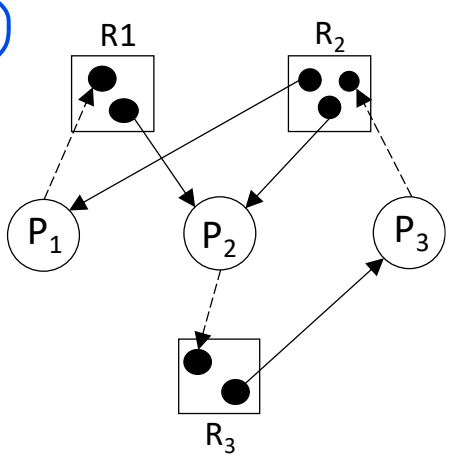
C.



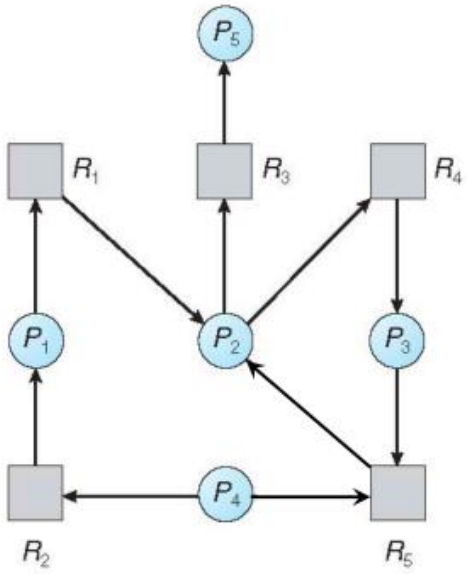
B.



D.



21. Consider a RAG that consists of several processes and instances of resources as illustrated in the following figure. Choose the **CORRECT** set(s) of the vertices for the RAG.



I. E = {R1 --> P2 , R2 --> P1 , R3 --> P5 , R4 --> P3 , R5 --> P2 }

II. E = {P1 --> R1 , P2 --> R3 , P2 --> R4 , P3 --> R5 , P4 --> R2 , P4 --> R5}

III. R = {R1, R2, R3, R4, R5}

IV. P = {P1, P2, P3, P4, P5}

A. I, II, III and IV

B. III and IV only

C. I and II only

D. None of the above